Ch.10 Table relationships

[Step 4]

A) Relationship type

B) Manner

C) Degree

Students

Student ID	StudFirst Name	StudLast Name	<< other fields >>
60001	Zachary	Ehrlich	.....
60002	Susan	Black	.....
60003	Joe	Rosales	.....
60004	Michael	Chow	.....
60005	Angie	Thompson	.....

Student Instruments

Student ID	Instrument ID	Checkout Date
60002	1000	09/26/11
60001	1002	09/28/11
60003	1010	09/28/11
60003	1013	09/28/11
60003	1011	09/28/11
60001	1022	10/02/11
60001	1021	10/02/11

Figure 10.1 *A pair of logically related tables*

- Importance:
  - Shows logical relations
  - Helps to further refine tables to eliminate redundancy
  - Allows joins between tables

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Part A) Types of relationship: 1-1

- Usually involves a subset table
- Q: why don't we combine the two tables into one?

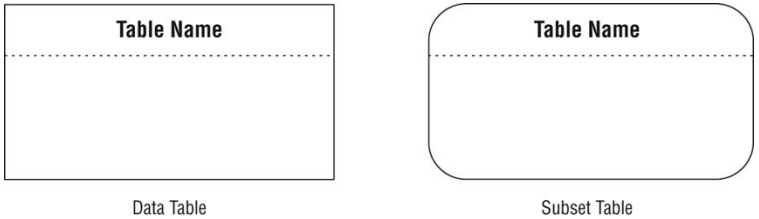


Figure 10.2 *Diagramming symbols for a data table and a subset table*

Employees

EmpID	EmpFirst Name	EmpLast Name	Home Phone	<< other fields >>
100	Zachary	Erlich	553-3992	.....
101	Susan	Black	790-3992	.....
102	Joe	Rosales	551-4993	.....

Compensation

EmpID	Hourly Rate	Commission Rate	<< other fields >>
100	25.00	5.0%	.....
101	19.75	3.5%	.....
102	22.50	5.0%	.....

Figure 10.4 *A typical example of a one-to-one relationship; Compensation is <u>not</u> a subset table*

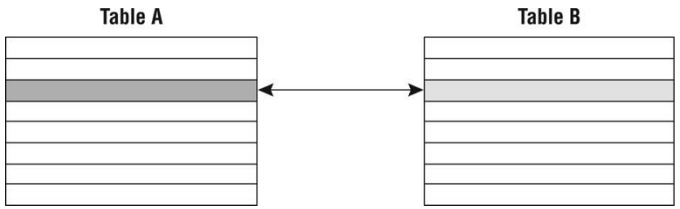


Figure 10.3 *A generic example of a one-to-one relationship*

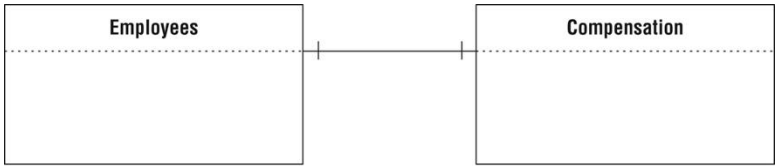


Figure 10.6 *The relationship diagram for the EMPLOYEES and COMPENSATION tables*

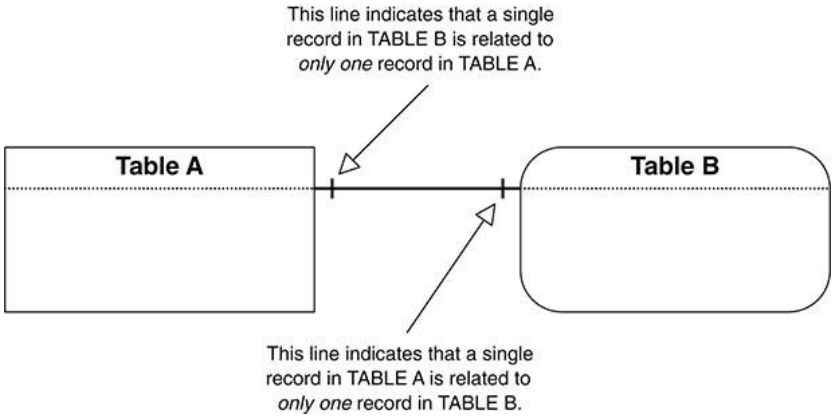


Figure 10.5 *Diagramming a one-to-one relationship*

Types of relationship: 1-N

- Most common

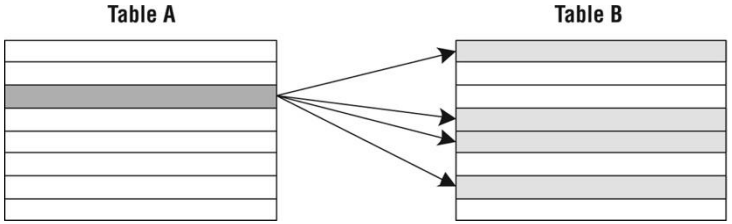


Figure 10.7 *A one-to-many relationship from the perspective of TABLE A*

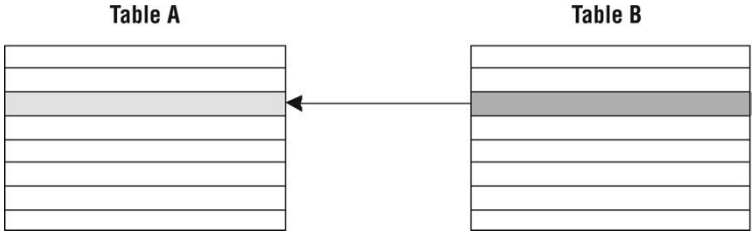


Figure 10.8 *A one-to-many relationship from the perspective of TABLE B*

Customers

Customer ID	CustFirst Name	CustLast Name	<< other fields >>
9001	Paul	Litwin	.....
9002	Alison	Balter	.....
9003	Andy	Baron	.....
9004	Chris	Kunicki	.....
9005	Mary	Chipman	.....

Customer Rentals

Customer ID	Item ID	Checkout Date
9002	80115	09/26/11
9001	64558	09/28/11
9003	10202	09/28/11
9003	11354	09/28/11
9003	78422	10/02/11
9005	30556	09/26/11
9004	20655	10/05/11

Figure 10.9 *A typical example of a one-to-many relationship*

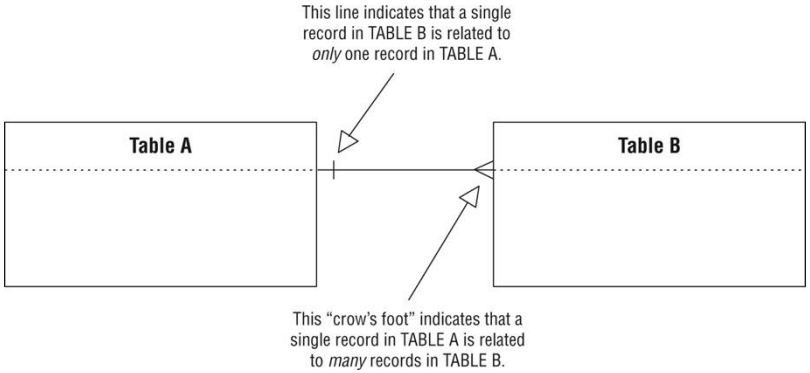


Figure 10.10 *Diagramming a one-to-many relationship*

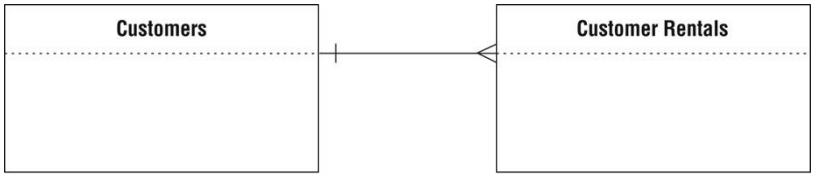


Figure 10.11 *The relationship diagram for the CUSTOMERS and CUSTOMER RENTALS tables*

Types of relationship: M-N

- Second most common

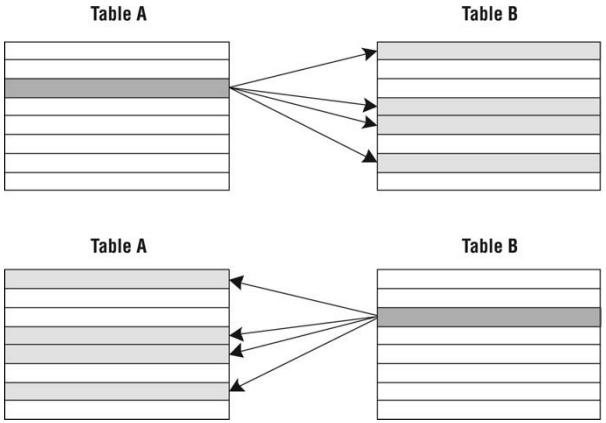


Figure 10.12 *A many-to-many relationship from the perspective of both TABLE A and TABLE B*

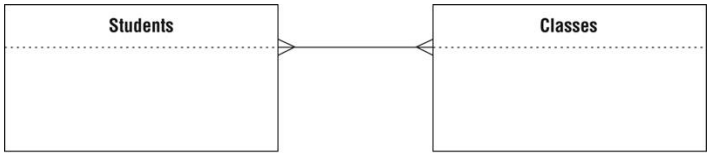
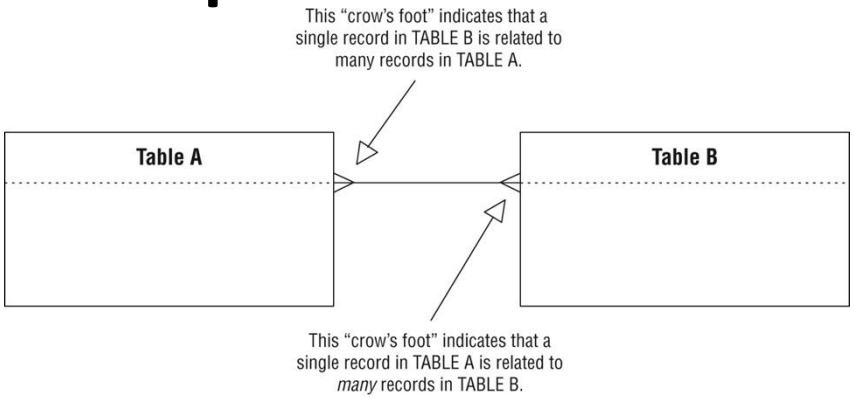


Figure 10.15 *The relationship diagram for the STUDENTS and CLASSES tables*

Figure 10.13 *A typical example of a many-to-many relationship*

Figure 10.14 *Diagramming a many-to-many relationship*



Students

Student ID	StudFirst Name	StudLast Name	StudStreet Address	StudCity	StudState	StudZipcode	<< other fields >>
60001	Zachary	Erlich	1204 Bryant Road	Seattle	WA	98125	.....
60002	Susan	Black	101 C Street, Apt. 32	Redmond	WA	98052	.....
60003	Joe	Rosales	201 Cherry Lane SE	Redmond	WA	98073	.....
60004	Diana	Price	4141 Lake City Way	Woodinville	WA	98072	.....
60005	Tom	Wickerath	2100 Mineola Avenue	Bellevue	WA	98006	.....

Classes

Class ID	Class Name	Class Category	Credits	Instructor ID	Classroom	<< other fields >>
900001	Advanced Calculus	Math	5	220087	2201	.....
900002	Advanced Music Theory	Music	3	220039	7012	.....
900003	American History	History	5	220148	3305	.....
900004	Computers in Business	Computer Science	2	220387	5115	.....
900005	Computers in Society	Computer Science	2	220387	5117	.....
900006	Introduction to Biology	Biology	5	220498	3112	.....
900007	Introduction to Database Design	Computer Science	5	220516	5105	.....
900008	Introduction to Physics	Physics	4	220087	2205	.....
900009	Introduction to Political Science	Political Science	5	220337	3308	.....

Self-referencing relationships: 1-1 and 1-N

Members

Member ID	MbrFirst Name	MbrLast Name	**Sponsor ID**	*<< other fields >>*
1001	Zachary	Erlich		.....
1002	Susan	Black	1001	.....
1003	Joe	Rosales		.....
1004	Diana	Price	1003	.....
1005	Tom	Wickerath	1002	.....

Figure 10.20 *Example of a self-referencing one-to-one relationship*

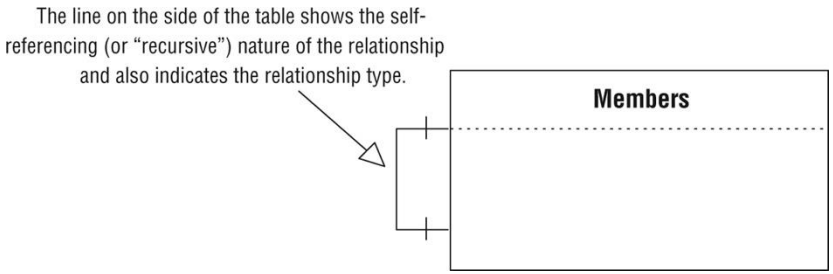


Figure 10.21 *Diagramming a self-referencing one-to-one relationship*

Customers

Customer ID	CustFirst Name	CustLast Name	**Referred By**	*<< other fields >>*
9001	Paul	Litwin		.....
9002	Alison	Balter		.....
9003	Andy	Baron	9001	.....
9004	Chris	Kunicki	9003	.....
9005	Mary	Chipman	9001	.....

Figure 10.22 *Example of a self-referencing one-to-many relationship*

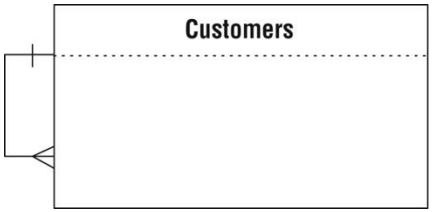


Figure 10.23 *Diagramming a self-referencing one-to-many relationship*

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Self-referencing relationships: M-N

Parts

Made of

Is part of

Part ID	Part Name	<< other fields >>
701	Top Clamp	.....
702	Bottom Clamp	.....
703	Fastening Bolt	.....
704	Clamp Assembly	.....
705	Saddle	.....
706	Seatpost	.....
707	Seat Assembly	.....
708	Body Tube	.....
709	Front Fork Tube	.....
710	Rear Stay Tube	.....
711	Frame Assembly	.....

Figure 10.24 *Example of a self-referencing many-to-many relationship*

Record 704 is 1-N with records (701, 702, 703)

Records (707, 711) are M-1 with record 704

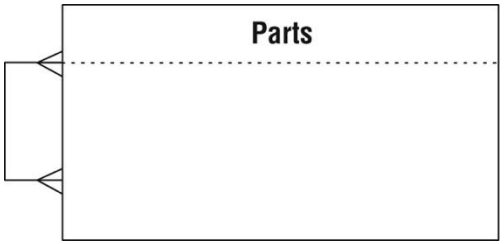

Figure 10.25 *Diagramming a self-referencing many-to-many relationship*

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	Buildings	Classes	Compensation	Faculty	Rooms	Staff	Students
Buildings							
Classes							
Compensation							
Faculty							
Rooms							
Staff							
Students							

Figure 10.26 *Setting up a table matrix to help identify existing relationships*

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Elements of a FK

1. Has the same name as the PK (only few exceptions, see alias example in the book; or if a FK is in the same table with a PK in 1-1 self-referencing table)
2. Is a replica of a PK (have the same name but they are in different tables; FK pulls values from PK)

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PK and its corresponding FK with different name, ok? No

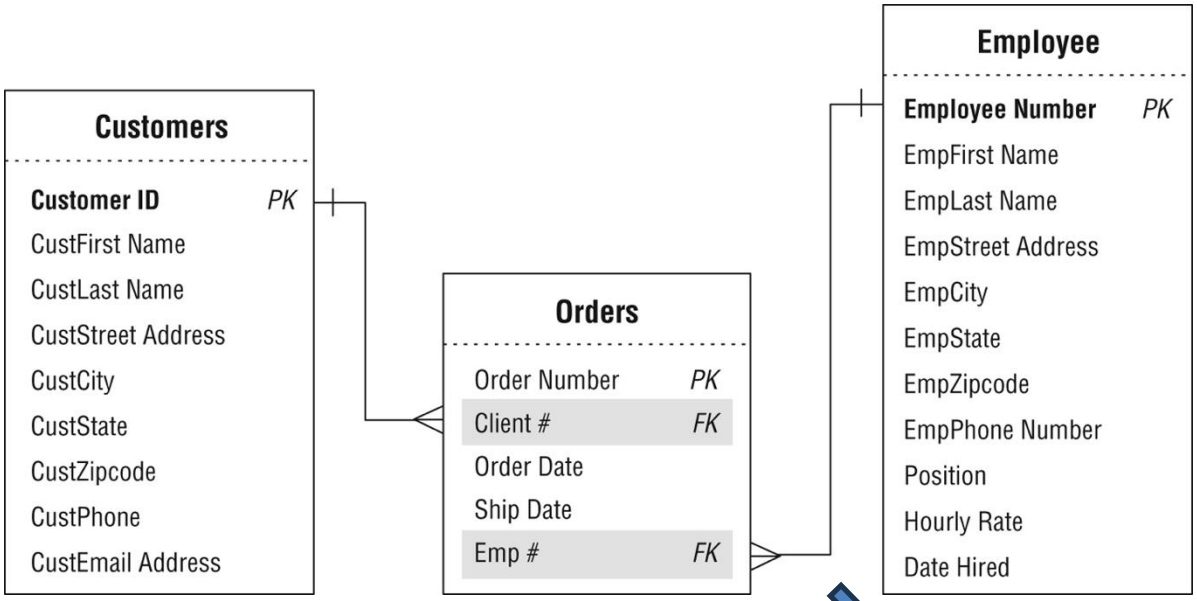

Figure 10.51 *Primary keys and foreign keys with mismatched names; brakes rule #1, is Emp# equivalent to Employee Number?*

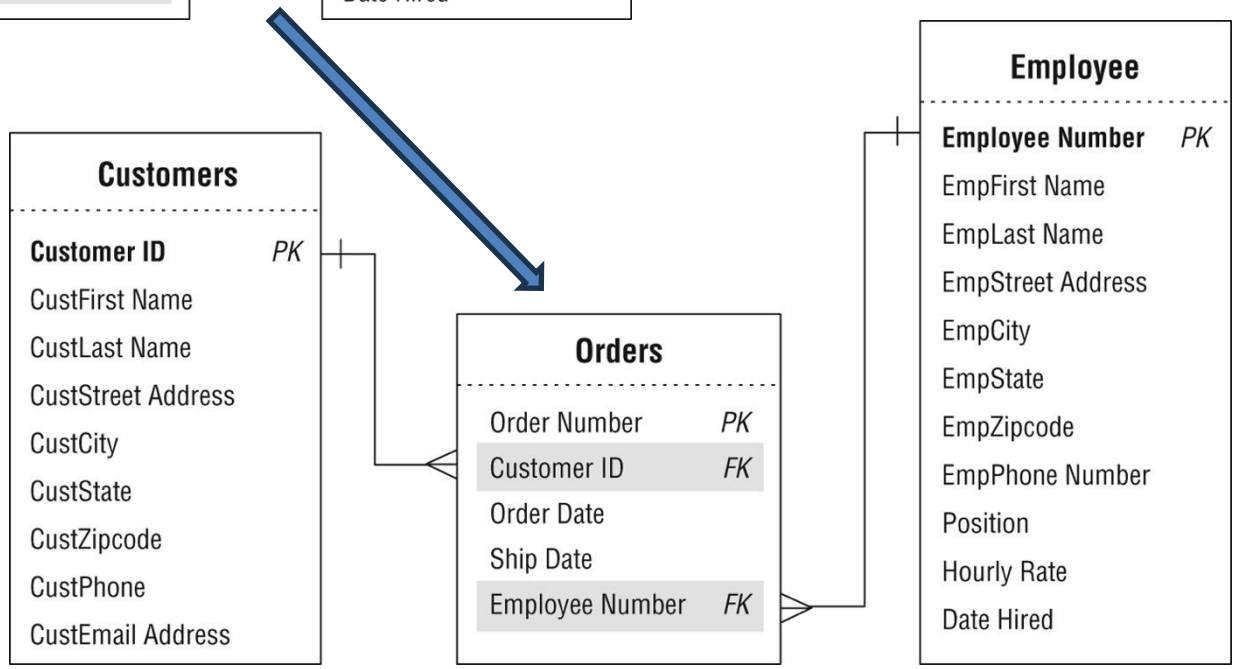

Figure 10.52 *Foreign keys that comply with the first element of a foreign key*

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Example: FK properties

General Elements

Field Name:	Employee ID Number	**Specification Type:**	☐ Unique ☐ Generic ☒ Replica
Parent Table:	Orders	**Source Specification:**	
Alias(es):		Employee ID Number from the Employees table.	
Shared By:			
Description:	The identification number of an employee within our organization. The values in this field enable us to identify and keep track of the employees who place orders for our customers.		

Figure 10.53 *General Elements for the* EMPLOYEE ID NUMBER *foreign key field in the ORDERS table*

Logical Elements

Key Type:	☐ Non	☐ Primary	**Edit Rule:**	
	☒ Foreign	☐ Alternate	☒ Enter Now, Edits Allowed	
Key Structure:	☒ Simple	☐ Composite	☐ Enter Now, Edits Not Allowed	
Uniqueness:	☒ Non-Unique	☐ Unique	☐ Enter Later, Edits Allowed	
Null Support:	☐ Nulls Allowed	☒ No Nulls	☐ Enter Later, Edits Not Allowed	
Values Entered By:	☒ User	☐ System	☐ Not Determined At This Time	
Required Value:	☐ No	☒ Yes		
Range of Values:	Any existing Employee ID Number in the Employees table.			

Note: usually, "No Nulls" and "Yes" Required Value go together; Required Value, means you cannot leave it blank, must write something there, even NULL.

Figure 10.54 *Logical Elements for the* EMPLOYEE ID NUMBER *foreign key field in the ORDERS table*

Logical Elements

Key Type:	☐ Non	☒ Primary	**Edit Rule:**	
	☐ Foreign	☐ Alternate	☐ Enter Now, Edits Allowed	
Key Structure:	☒ Simple	☐ Composite	☒ Enter Now, Edits Not Allowed	
Uniqueness:	☐ Non-Unique	☒ Unique	☐ Enter Later, Edits Allowed	
Null Support:	☐ Nulls Allowed	☒ No Nulls	☐ Enter Later, Edits Not Allowed	
Values Entered By:	☐ User	☒ System	☐ Not Determined At This Time	
Required Value:	☐ No	☒ Yes		
Range of Values:	1000 - 9999			

Figure 10.55 *Logical Elements for the* EMPLOYEE ID NUMBER *primary key field in the EMPLOYEES table*

Deletion rules for each relationship – **from parent perspective!**

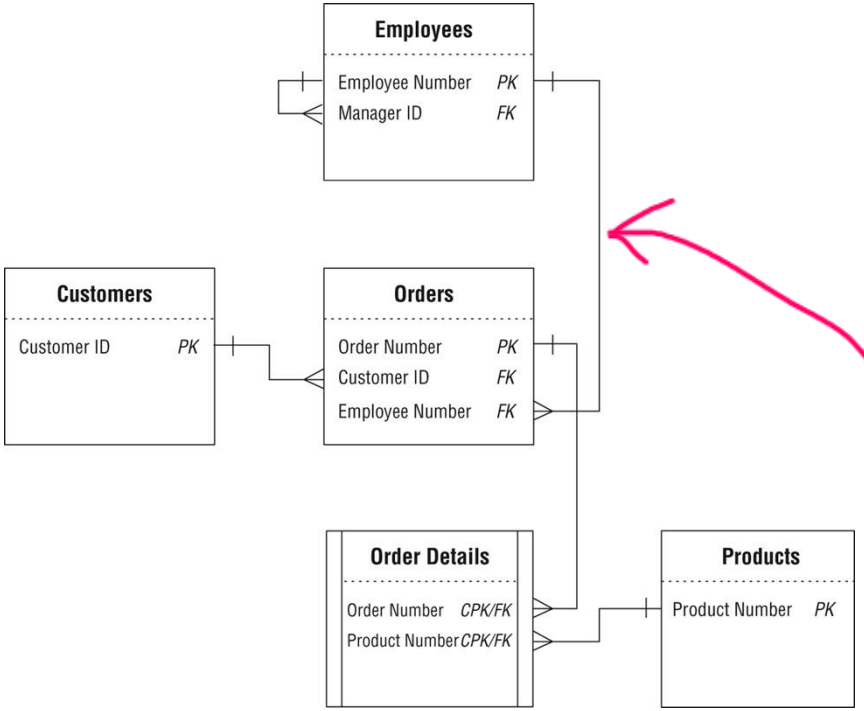

Figure 10.56 *What deletion rule is appropriate for a given relationship?*

Q: When a record in the EMPLOYEES table is deleted, what should happen to related records in the ORDERS table?

A: Can't delete an employee record; have to designate the employee as inactive. ➔ ***Deny***

A: Must first delete the orders associated with the employee from the ORDERS table and then delete the employee from the EMPLOYEES table ➔ ***Restrict***

A: All orders associated with the employee must be deleted from the ORDERS table as well ➔ ***Cascade***

A: Employee number for all orders associated with the employee must be reset to Null ➔ ***Nullify***

A: Employee number for all orders associated with the employee must be reset to the lead salesperson's employee number➔ ***Set Default***

1. Deny – no delete, label record "inactive" in parent table
2. Restrict – delete records in parent table only after all related records in child table are deleted
3. Cascade – delete all records in parent table and all related records in child table
4. Nullify - delete records in parent table and update the FK in related table to NULL (check "Nulls allowed" in FK specifications)
5. Set Default - delete records in parent table and update the FK in related table to a default value (must have it in FK specifications)

Deletion rules example

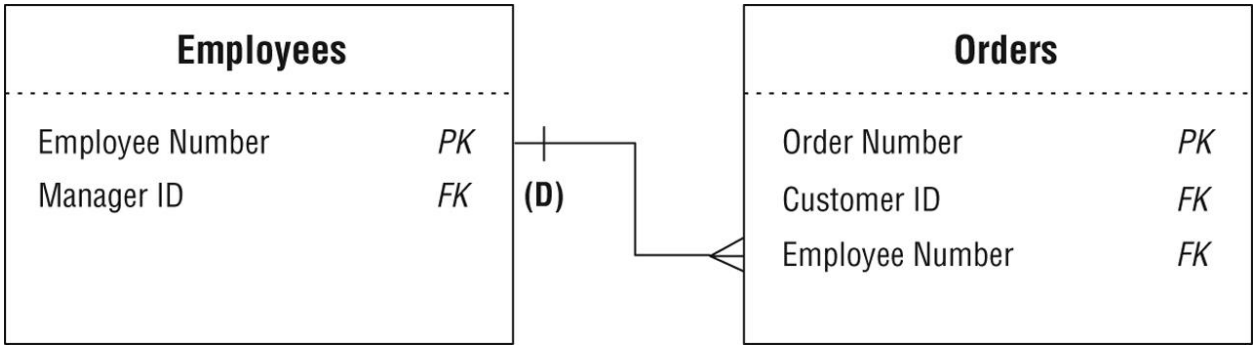

Figure 10.57 *Designating a Deny deletion rule for the relationship between the EMPLOYEES and ORDERS tables*

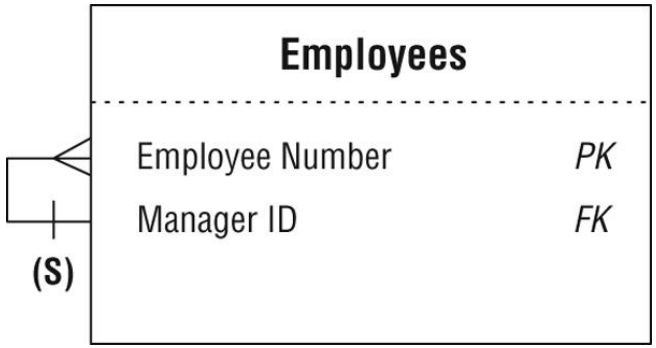

Figure 10.58 *Designating a Set Default deletion rule for the EMPLOYEES table self-referencing relationship*
If an employee leaves the corporation, you replace that Manager ID entries with the default of the Director.

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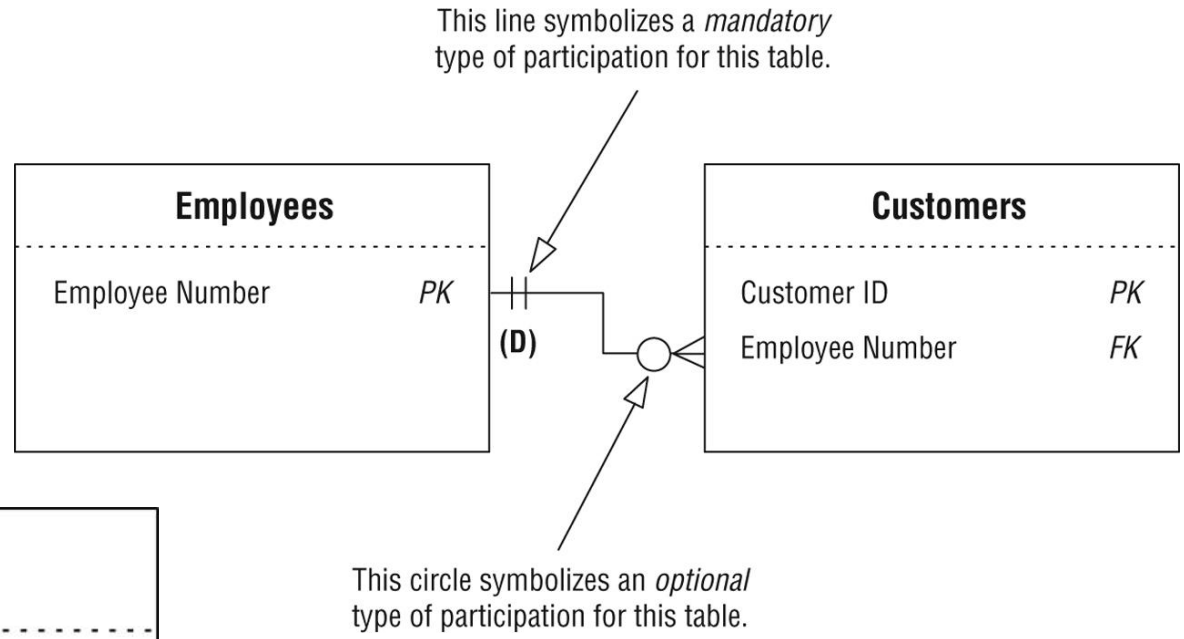
Part B) Manner of participation

1. Mandatory – at least one record must exist in parent before entering one in child table
2. Optional

Figure 10.60 *Designating the type of participation for the EMPLOYEES and CUSTOMERS tables*



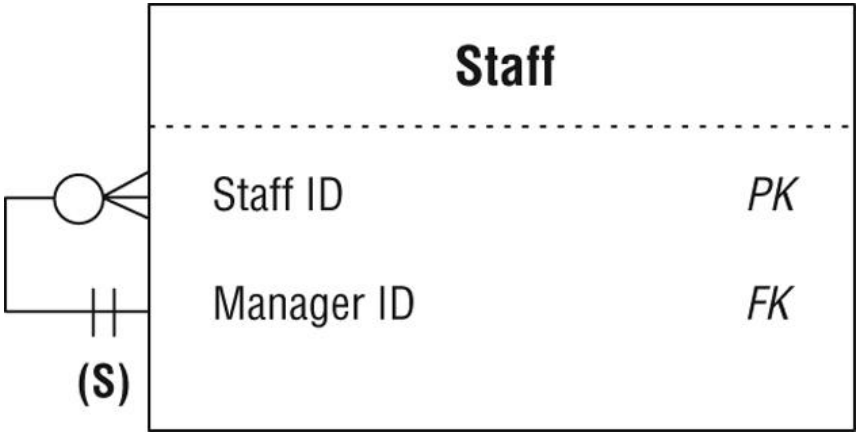


Figure 10.61 *Designating the type of participation for the primary and foreign keys of the STAFF table; is this correct? yes*

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Part C) Degree of participation

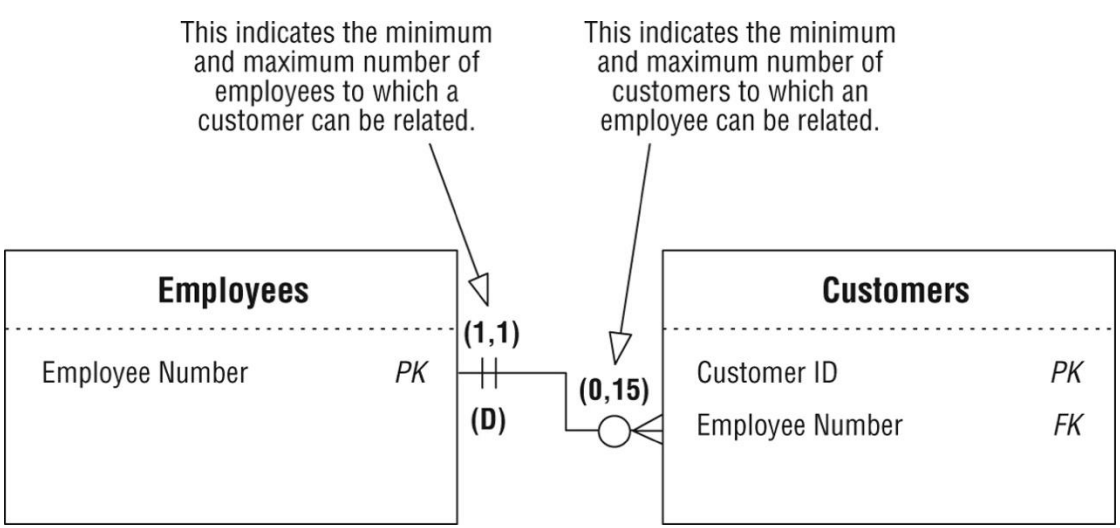


Figure 10.62 *Designating the degree of participation for the EMPLOYEES and CUSTOMERS tables*

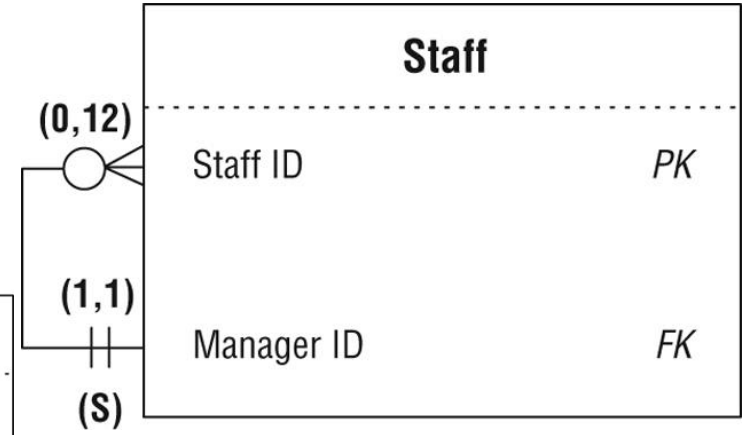
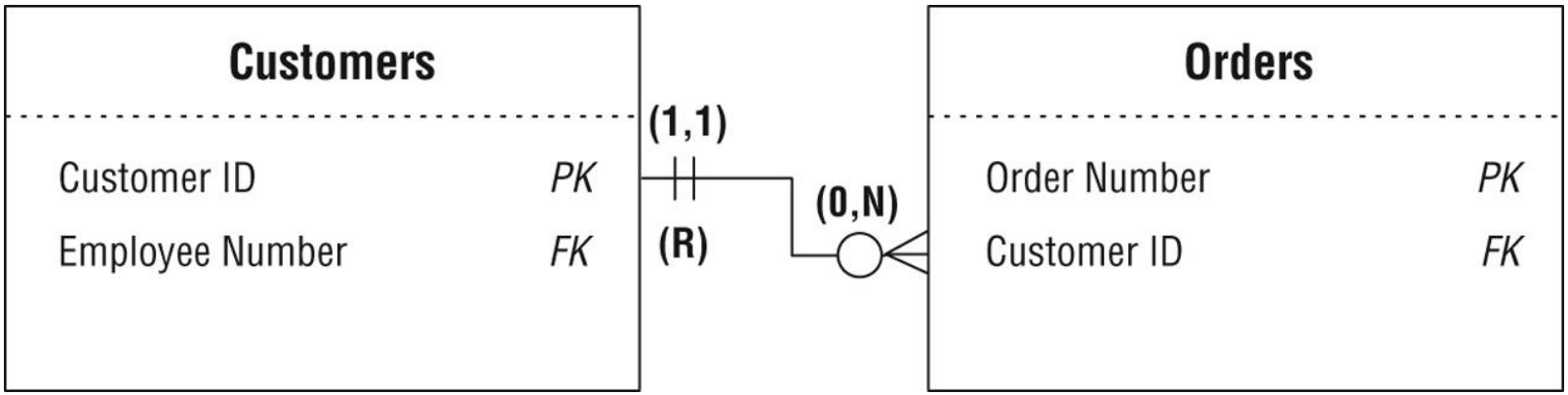


Figure 10.63 *Designating the degree of participation for the primary and foreign keys of the STAFF table*

Figure 10.64 *Designating an unlimited degree of participation for the ORDERS table*



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